

ZEPTO SALES ANALYSIS

Data Cleaning, Dashboarding & Business Insights

A quick-commerce sales analytics project

Project Report
Version 1.0
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1. Project Overview

My project analyzes 1,500 quick-commerce order records from a Zepto-style delivery operation spanning three cities — Bangalore, Delhi, and Mumbai — over a January–March 2025 window. The raw export arrived in the state such data almost always arrives in: inconsistent category and payment-method labels, three different date formats in a single column, a currency symbol mixed into a price field, missing location values, and one unlabelled column with no documented meaning.

The project takes that raw file through a complete, small-scale analytics workflow: profiling the data to understand exactly what was wrong with it, cleaning it into a single trustworthy table, building an interactive dashboard on top of the cleaned data, and extracting the business insights that dashboard makes visible. The end result is not just a set of charts — it is a documented, repeatable path from a messy spreadsheet to numbers a business stakeholder could act on with confidence.

The scope deliberately stays close to what the data actually supports. Every chart and every insight in this report is computed directly from the 1,500 records in the source file after cleaning; nothing here is illustrative or assumed. Where the data has a real limitation — for example, March 2025 having only two days of orders — that limitation is called out rather than smoothed over.

2. Objective

The project has one overarching goal: turn an unreliable raw sales export into a dataset and a dashboard that can be trusted for day-to-day operational decisions. That goal breaks down into four concrete objectives:

- Identify and fix every data quality issue in the raw file — inconsistent categories, mixed date formats, embedded currency symbols, and missing values — without silently discarding or guessing at information the data does not actually give.
- Build a single, analysis-ready dataset where every numeric and categorical field can be trusted to mean what its column header says, and where the relationship between fields (e.g., how Total is calculated) is confirmed rather than assumed.
- Design a dashboard that surfaces the metrics a quick-commerce operations or category team would actually check day to day: revenue, delivery performance, category mix, payment mix, and warehouse-level performance.
- Translate what the cleaned data and dashboard show into a small set of specific, numbers-backed insights and, where relevant, the operational question each insight raises.

- These objectives are ordered deliberately: cleaning has to happen before the dashboard can be trusted, and the dashboard has to exist before the insights can be pulled from it rather than asserted.

3. Dataset Information

Attribute	Detail
Source file	1784395712986_zepto_sales_raw.xlsx
Records	1,500 orders
Fields	28 columns (order, customer, product, pricing, delivery, and outcome data)
Time span	January 1, 2025 – March 1, 2025
Geography	Bangalore, Delhi, Mumbai
Grain	One row per order

The dataset covers the full order lifecycle: what was ordered (SKU, product, category, quantity, price), who ordered it (customer type, segment, city), how it was paid for, how it was fulfilled (warehouse, rider, delivery slot, delivery time), and what happened to it (delivered, cancelled, failed, or returned, with a reason code and any refund). This breadth is what makes the dataset suitable for a dashboard rather than a single report — it supports revenue, delivery-performance, and operational views from the same table.

4. Tools Used

Tool	Purpose
Excel / .xlsx	Source data format and raw file inspection
Power query(Excel)	Importing, cleaning, and transforming raw data
Pivot tables	Summarizing and aggregating sales data
Pivot charts	Visualizing summarized data
Slicers	Creating interactive dashboard filters
Microsoft Word	This project report

5. Data Cleaning

Cleaning was driven directly by the issues found during profiling of the raw file, rather than a generic checklist. Each fix below states the problem, the rule applied, and how many of the 1,500 rows it affected, so the cleaning is auditable rather than a black box.

5.1 Price — Currency Symbol Removal

226 of 1,500 rows stored Price with a ₹ prefix (e.g., "₹128.14"), which forced the entire column into a text type and made it unusable in arithmetic. The ₹ symbol was stripped from every value and the column was cast to a numeric type. After conversion, Price × Qty − Discount was recomputed and checked against the existing Subtotal column for all 1,500 rows — it matched exactly, confirming the fix did not distort the underlying pricing logic.

5.2 Order_Date — Multi-Format Parsing

Order_Date mixed three formats within the same column: ISO YYYY-MM-DD (529 rows), DD/MM/YYYY (470 rows), and "DD Mon YYYY" text such as "15 Jan 2025" (501 rows). A format-aware parser tried each pattern in turn for every value and converted the column to a single true date type. All 1,500 values parsed successfully with zero failures, and the resulting date range (January 1 – March 1, 2025) was verified as internally consistent with the rest of the dataset.

5.3 Category — Normalization

The raw Category field held 16 distinct text values for what are really 10 categories, caused by inconsistent casing ("frozen" vs "Frozen"), abbreviation ("snaks" vs "Snacks"), and outright misspelling ("bevrage" for Beverages, "DIARY" for Dairy). Every value was lowercased, trimmed, and mapped through a reference lookup table to one canonical label per category. This consolidation changed category-level totals materially — for example, what looked like two mid-sized categories ("Beverages" and "bevrage") is really one larger category once merged.

5.4 Payment_Method — Normalization

Payment_Method held 9 distinct values for 7 real payment channels, primarily driven by inconsistent casing and spacing in how UPI was recorded ("UPI", "upi", "UP I"). These were merged into a single "UPI" label using the same normalization approach as Category. UPI is, after correction, the dominant payment method by a wide margin rather than one of several similarly sized options — a conclusion the raw, unmerged data would have hidden.

5.5 Missing Values — Rule-Based Handling

Rather than applying one missing-data rule to the whole file, each field's missingness was checked against the rest of the row before deciding how to treat it:

- Reason (85.4% missing) and Delivery_Time (16.3% missing) were confirmed to be missing only, and exactly, when Delivery_Status made that expected — Reason is populated only for non-Delivered orders, Delivery_Time only for Delivered ones. These nulls were left as-is; imputing either would have invented information the business process never recorded.
- Pincode (69 rows) and City (28 rows) had no such explanation — their missingness did not correlate with any other field. These rows were retained in the cleaned dataset but flagged, and excluded specifically from city- and location-level aggregations rather than dropped from the dataset entirely.

5.6 Record-Level Anomalies

Three orders (O0580, O0745, O0992) recorded a Delivery_Time of 0 minutes while marked "Delivered," which is not physically plausible. These three rows were flagged rather than corrected, since the true delivery time cannot be recovered from the data available, and were excluded only from the average-delivery-time calculation reported in Section 7.

5.7 Result

The cleaning process produced a single analytical table of 1,500 rows with standardized categories, payment methods, and a true date type; a numeric, arithmetic-ready Price field; and every remaining null explained rather than silently filled. This is the table the dashboard in Section 6 and the insights in Section 7 are built from.

6. Dashboard Features

The dashboard is a single-page interactive view built on the cleaned dataset, organized around the questions an operations or category manager checks most often: how much revenue is coming in, from where, through which categories and payment methods, and how reliably orders are being fulfilled.

6.1 KPI Summary Row

Eight headline metrics are shown as always-visible cards at the top of the dashboard: total orders, total revenue, net revenue (after refunds), average order value, delivered percentage, average delivery time, total refunds, and substitution rate. These give a one-glance health check before drilling into any chart.

KPI Cards: Displays key business metrics such as Total Sales, Total Profit, Total Orders, Quantity Sold, and Average Sales for quick performance evaluation.

6.2 Revenue and Category Views

- Revenue by Category — a horizontal bar chart ranking all 10 (cleaned) categories by revenue contribution.
- Revenue by City — a bar chart comparing Bangalore, Delhi, and Mumbai
- Monthly Revenue Trend — a line chart showing revenue across the three months covered by the data.
- Top products — the highest-revenue individual SKUs, available as a supporting data view.

6.3 Delivery and Operations Views

- Orders by Delivery Outcome — a donut chart showing the Delivered / Cancelled / Failed / Returned split.
- Cancellation / Failure / Return Reasons — a bar chart breaking down why orders did not complete successfully.
- Warehouse Performance table — order volume and revenue for each of the 7 warehouses, sortable by revenue, to support fulfilment-side comparison.

6.4 Customer and Payment Views

- Payment Method Mix — a pie chart of the 7 cleaned payment channels.
- Customer segment revenue — revenue broken out by the six business segments served (Office, Café, Tea Stall, Household, Bakery, Cloud Kitchen).

All charts are computed live from the same cleaned, aggregated dataset, so every number in the dashboard traces back to the cleaning rules in Section 5 rather than to a separately maintained summary.

6.5 Slicers

Interactive Filters (Slicers): Enables users to filter dashboard data by Region, Category, Product, Month, and Year for customized analysis.

For example, users can select a specific **Region** to view sales performance within that geographical area, filter by **Product Category** to compare category-wise revenue and profit, or choose a particular **Month** or **Year** to analyze seasonal trends and year-over-year performance.

Multiple slicers can also be used simultaneously, enabling more detailed analysis, such as viewing the sales of a specific product category within a selected region during a particular time period.

7. Key Insights

7.1 Revenue Performance

Total revenue across all 1,500 orders is ₹563,992, with net revenue of ₹491,230 after ₹72,763 in refunds — meaning refunds account for roughly 12.9% of gross revenue, a level worth monitoring given it is concentrated almost entirely in non-delivered orders (Section 7.3). Average order value is ₹375.99, and Bangalore is the strongest city both in order volume (516 orders) and revenue (₹201,866), ahead of Delhi (₹180,571) and Mumbai (₹169,384).

7.2 Category Mix

Beverages is the single largest category by revenue (₹82,603) once the "beverage" misspelling is merged in, followed by Frozen (₹76,044), Veg (₹73,582), and Snacks (₹72,713). The bottom of the ranking — Baby (₹34,274) and Bakery (₹29,579) — generate under half the revenue of the top category, suggesting the assortment is comparatively narrow-focused around a handful of high-velocity categories rather than evenly spread across all ten.

7.3 Delivery Performance

83.7% of orders were successfully delivered; the remaining 16.3% split across Cancelled (7.6%), Failed (5.5%), and Returned (3.2%). Among orders that did not complete, Stockout (47 cases) and Delay (44 cases) are the two leading reasons, ahead of Customer Cancel (40), Damaged (36), Payment Failure (28), and Wrong Item (24) — pointing toward inventory availability and on-time fulfilment as the two biggest levers for reducing the 16.3% non-delivery rate, ahead of payment or last-mile packaging issues. Average delivery time for successfully delivered orders is 24.7 minutes, and 61.5% of orders are fulfilled within the fastest 10–20-minute slot.

7.4 Payment Behaviour

After normalization, UPI is the dominant payment method at 493 orders (32.9% of all orders) — nearly triple the volume of the next-largest single method once its three raw spellings are correctly merged. NetBanking (186), GPay (182), Card (165), PhonePe (161), Wallet (158), and COD (155) form a comparatively even second tier, together accounting for the remaining two-thirds of volume.

7.5 Warehouse and Fulfilment

Koramangala_02 generates the highest single-warehouse revenue (₹88,431 across 232 orders), closely followed by Saket_01 (₹87,297) and Andheri_02 (₹87,082); Bandra_01 is the lowest-revenue warehouse (₹63,885 across 182 orders). Revenue differences track order volume closely across warehouses, which suggests no single warehouse is meaningfully under- or over-

performing on order value once volume is accounted for — the gap is one of demand, not efficiency.

7.6 Substitutions and Customer Segments

48.5% of all orders involved at least one item substitution — nearly one in two orders — which is high enough to be a customer-experience concern worth its own investigation, even though it was outside this project's scope to determine which specific SKUs drive it. By customer segment, Office (₹103,708) is the highest-revenue segment, followed closely by Café (₹96,039) and Tea Stall (₹94,848); the six segments are otherwise fairly evenly matched, indicating the business is not overly dependent on any single customer segment.

8. Conclusion

The Interactive Sales Dashboard successfully transforms raw sales data into meaningful business insights through effective data analysis and visualization. By leveraging Microsoft Excel / Power BI, the dashboard enables users to monitor key performance indicators, analyze sales trends, evaluate regional and product performance, and make data-driven decisions. This project demonstrates practical skills in data cleaning, dashboard development, KPI reporting, and business intelligence, making it a valuable solution for efficient sales performance analysis.